Performance Benchmarking of ROS 2 DDS Middleware for High-Frequency Multi-Agent Coordination in Industrial IoT

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EXTENDED ABSTRACT

As multi-robot systems scale in industrial environments, the underlying DDS (Data Distribution Service) middleware dictates communication determinism and latency. This paper presents an exhaustive empirical benchmark comparing Eclipse CycloneDDS and eProsima FastDDS across varied network configurations, message payload sizes, and QoS profiles. The authors quantify latency, packet jitter, and throughput for ROS 2 Topics, Services, and Actions under artificial packet drop conditions.

Keywords: ROS 2, DDS Benchmarking, CycloneDDS, FastDDS, Quality of Service (QoS), Multi-Agent Systems

1. Industrial Middleware Requirements

In heterogeneous workcells, communication failure during an Action feedback loop or Service call can cause catastrophic timing misalignments. The paper investigates how different DDS vendor implementations handle multi-node discovery and high-frequency message streaming.

2. QoS Configuration Profiles for Multi-Robot Cells

The paper recommends specific Quality of Service policies for heterogeneous robotics:

- Action Goals & Results: Reliability = RELIABLE, Durability = TRANSIENT_LOCAL, History = KEEP_ALL.
- High-Frequency Feedback (10-50 Hz): Reliability = BEST_EFFORT, History = KEEP_LAST (depth=5).
- Mutual Exclusion Services: Reliability = RELIABLE with explicit TCP/UDP unicast fallbacks.

3. Latency Benchmarks Summary

Primitive Type	FastDDS Latency (ms)	CycloneDDS Latency (ms)	Jitter (ms)
Topics (64 bytes, 100 Hz)	0.85 ms	0.62 ms	±0.12 ms
Services (Request-Response)	12.8 ms	11.1 ms	±0.45 ms
Actions (Goal to Feedback)	18.5 ms	16.2 ms	±1.10 ms

DIRECT RELEVANCE & SYNTHESIS FOR TEAM 3 PROJECT

Establishes the communication layer and QoS settings required in our multi-robot launch files, ensuring sub-15 ms deterministic service handshakes between UR5 and TurtleBot3.